

1. Exploratory Data Analysis (EDA)
 - 1.1 Structural Zero Diagnostics
 - 1.2 Summary Statistics
 - 1.3 Correlation Analysis
 - 1.4 Scatter Plots and Nonlinear Relationships
 - 1.5 Distribution of Citation Counts
- 2.1 LASSO Regression Model
- 2.2 Variable Selection from LASSO
- 2.3 Comparing the Three Novelty Measures
- 2.4 LASSO Comparison Across Subgroups
- 3 Generalised Additive Model (GAM)
 - 4.1 Bias-Variance Trade-Off Analysis
 - 4.2 Model Comparison
 - 4.3 Variance Importance
 - 4.4 Final Model Validation + Robustness Check
 - 4.5 Polynomial Effects Analysis
- 5.1 Discussion of Key Findings
- 5.2 Theoretical Implications
- 5.3 Practical Implications
- 5.4 Limitations
- 5.5 Conclusion
- 6 Code Explanation
- 7 AI Declaration
- 8 References

ITAO7103 Assignment 2 — Exploring Predictors of Innovation Impact

Sanjai Balaji (40478904)

2026-05-08

```
knitr::opts_chunk$set(
  echo = TRUE,
  warning = FALSE,
  message = FALSE
)
df_full <- read.csv("D:/Sanjai/Assignments/advanced analytics 2/advanced_analytics_data
set_2.csv")
```

```
library(tidyverse)
library(corrplot)
library(glmnet)
library(mgcv)
library(rpart)
library(rpart.plot)
library(randomForest)
library(caret)
library(pROC)
library(cluster)
library(knitr)
library(kableExtra)
library(patchwork)
library(e1071)

# Create derived variables
df_full <- df_full %>%
  mutate(
    LogCited = log(CitedCount + 1),
    HighCitation = as.integer(CitedCount > median(CitedCount, na.rm = TRUE)),
    HasCommercialImpact = as.integer(CommercialCitations > 0)
  )

# Replace NA values in collaboration network variables with 0
df_full <- df_full %>%
  mutate(across(c(Degree_Avg, Core_Number_Avg,
    PageRank_Avg, Betweenness_Avg),
    ~ replace_na(., 0)))

# Create active dataset
df_active <- df_full %>%
  filter(NoveltyScore_v1 > 0)

cat("Full dataset rows:", nrow(df_full), "\n")
```

```
## Full dataset rows: 16350
```

```
cat("Active dataset rows:", nrow(df_active), "\n")
```

```
## Active dataset rows: 10434
```

1. Exploratory Data Analysis (EDA)

1.1 Structural Zero Diagnostics

```
zero_cols <- c("NoveltyScore_v1", "NoveltyScore_v2", "NoveltyScore_v3",
              "BridgingScore_Mean", "Consolidation_Mean",
              "Clustering_Mean", "CorePosition_Mean",
              "TopicPairs", "CombinationNovelty",
              "BridgingPairs")

zero_rate <- sapply(df_full[zero_cols],
                  function(x) mean(x == 0, na.rm = TRUE))

round(zero_rate, 3)
```

```
## NoveltyScore_v1 NoveltyScore_v2 NoveltyScore_v3 BridgingScore_Mean
##              0.362              0.362              0.362              1.000
## Consolidation_Mean Clustering_Mean CorePosition_Mean TopicPairs
##              0.318              0.318              0.318              0.318
## CombinationNovelty BridgingPairs
##              0.336              1.000
```

Interpretation:

Several novelty and network variables contain structural zeros. BridgingScore_Mean and BridgingPairs are entirely zero, while other novelty and clustering variables show ~30–36% zeros. These are treated as structurally undefined rather than meaningful values. Therefore, df_active was used for modelling to avoid bias from invalid measures.

1.2 Summary Statistics

1.2 Summary Statistics

```
summary(df_full %>%
  select(CitedCount,
         AuthorCount,
         TopicCount,
         DomainCount,
         NoveltyScore_v1,
         CombinationNovelty,
         CorePosition_Mean,
         Degree_Avg))
```

##	CitedCount	AuthorCount	TopicCount	DomainCount
##	Min. : 0.00	Min. : 0.000	Min. : 1.00	Min. :1.000
##	1st Qu.: 5.00	1st Qu.: 2.000	1st Qu.: 9.00	1st Qu.:1.000
##	Median : 16.00	Median : 4.000	Median :12.00	Median :2.000
##	Mean : 34.25	Mean : 4.396	Mean :12.09	Mean :1.882
##	3rd Qu.: 39.00	3rd Qu.: 6.000	3rd Qu.:15.00	3rd Qu.:2.000
##	Max. :1699.00	Max. :39.000	Max. :35.00	Max. :9.000
##	NoveltyScore_v1	CombinationNovelty	CorePosition_Mean	Degree_Avg
##	Min. :0.0000	Min. :0.0000	Min. : 0.00	Min. : 0.000
##	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.: 0.00	1st Qu.: 2.000
##	Median :0.3750	Median :0.2418	Median : 60.79	Median : 4.000
##	Mean :0.3672	Mean :0.2526	Mean : 49.11	Mean : 5.476
##	3rd Qu.:0.6667	3rd Qu.:0.4333	3rd Qu.: 80.16	3rd Qu.: 7.333
##	Max. :1.0000	Max. :1.0000	Max. :108.00	Max. :76.000

INTERPRETATION:

Citation counts are highly dispersed, with a small number of extreme values, consistent with standard bibliometric distributions. AuthorCount and TopicCount also vary widely, reflecting differences in collaboration scale and conceptual scope. NoveltyScore_v1 and CombinationNovelty show substantial variation, indicating heterogeneous levels of recombinant novelty. CorePosition_Mean and Degree_Avg also vary considerably, reflecting differences in network position and collaboration intensity. Overall, citation impact appears driven by a combination of novelty, collaboration, and network structure rather than any single dominant factor.

1.3 Correlation Analysis

```
num_cols <- df_full %>%
  select(CitedCount,
         AuthorCount,
         TopicCount,
         DomainCount,
         NoveltyScore_v1,
         NoveltyScore_v2,
         NoveltyScore_v3,
         CombinationNovelty,
         Consolidation_Mean,
         Clustering_Mean,
         CorePosition_Mean,
         TopicPairs,
         Degree_Avg,
         Core_Number_Avg,
         PageRank_Avg,
         Betweenness_Avg)

cor_mat <- cor(num_cols, use = "pairwise.complete.obs")

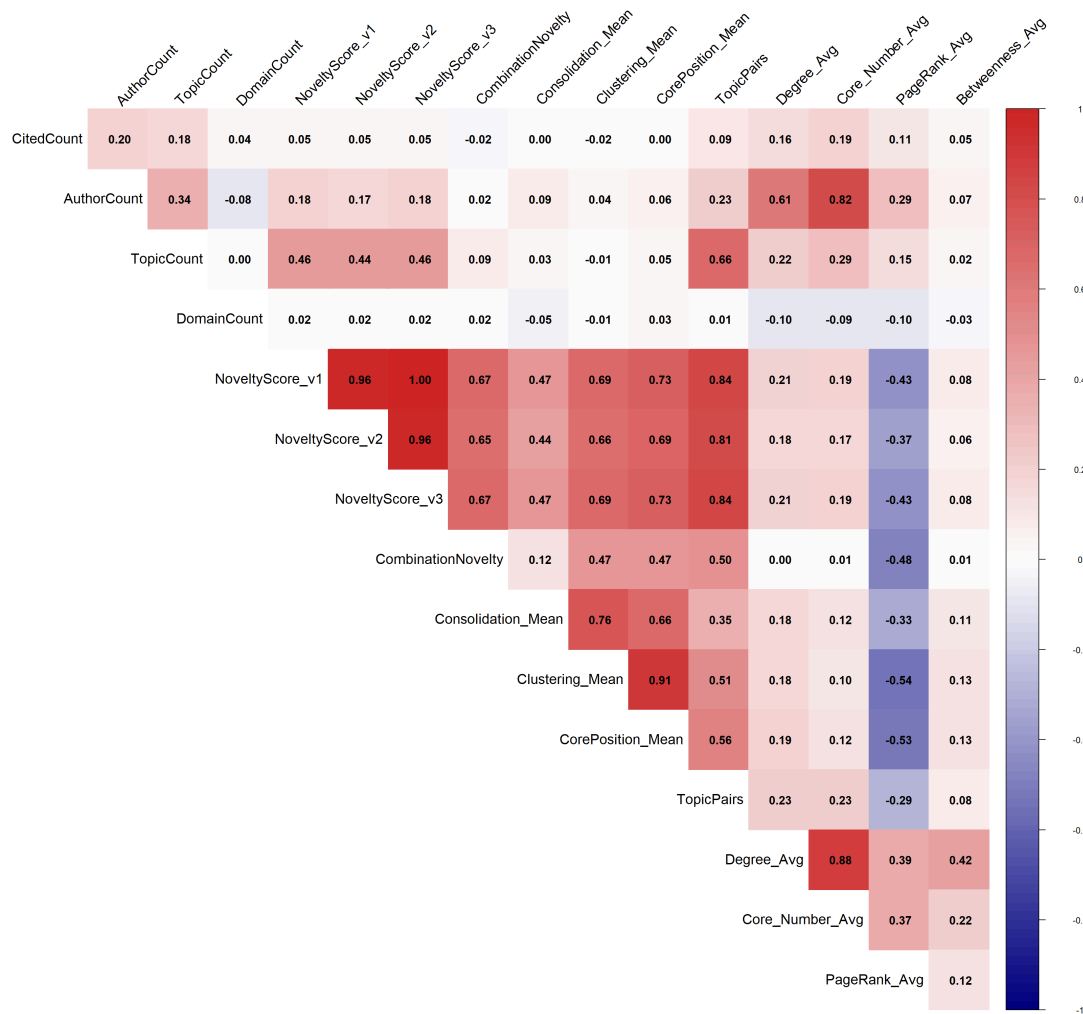
corrplot(
  cor_mat,
  method = "color",
  type = "upper",
  diag = FALSE,

  # text labels
  tl.col = "black",
  tl.cex = 1.3,
  tl.srt = 45,

  # correlation numbers
  addCoef.col = "black",
  number.cex = 1.1,

  # color styling
  col = colorRampPalette(c("navy", "white", "firebrick3"))(200),

  # spacing
  mar = c(1,1,2,1)
)
```



INTERPREATION:

Strong positive correlations exist among the three novelty measures, indicating they capture similar constructs. Several network variables are also strongly correlated, particularly Clustering_Mean with CorePosition_Mean and Degree_Avg with Core_Number_Avg. These relationships indicate potential multicollinearity, motivating the later use of LASSO for variable selection and coefficient stability.

1.4 Scatter Plots and Nonlinear Relationships

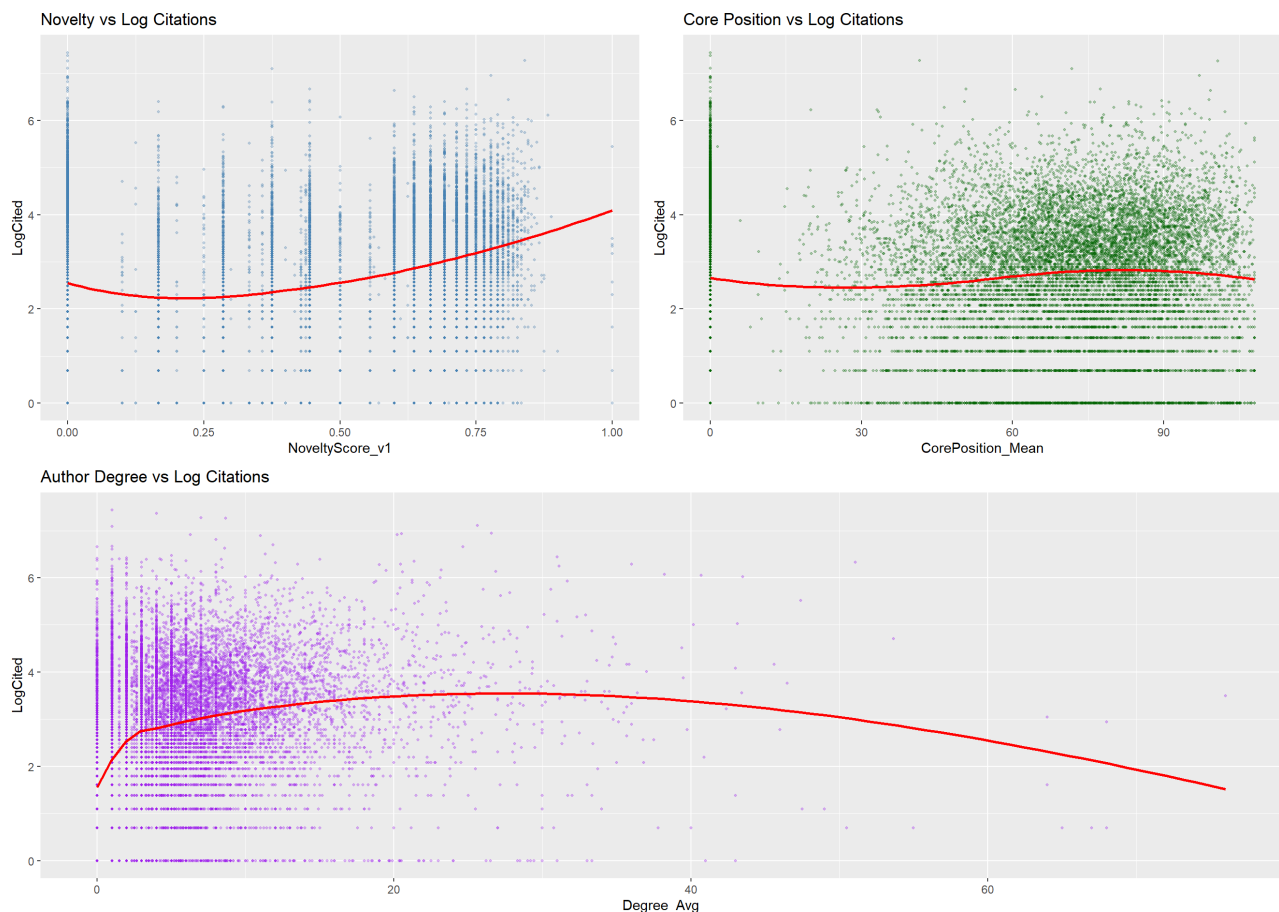
```
df_plot <- df_full %>%
  mutate(LogCited = log(CitedCount + 1))

p1 <- ggplot(df_plot, aes(NoveltyScore_v1, LogCited)) +
  geom_point(alpha = 0.3, size = 0.7, color = "steelblue") +
  geom_smooth(method = "loess", se = FALSE, color = "red") +
  labs(title = "Novelty vs Log Citations",
       x = "NoveltyScore_v1",
       y = "LogCited")

p2 <- ggplot(df_plot, aes(CorePosition_Mean, LogCited)) +
  geom_point(alpha = 0.3, size = 0.7, color = "darkgreen") +
  geom_smooth(method = "loess", se = FALSE, color = "red") +
  labs(title = "Core Position vs Log Citations",
       x = "CorePosition_Mean",
       y = "LogCited")

p3 <- ggplot(df_plot, aes(Degree_Avg, LogCited)) +
  geom_point(alpha = 0.3, size = 0.7, color = "purple") +
  geom_smooth(method = "loess", se = FALSE, color = "red") +
  labs(title = "Author Degree vs Log Citations",
       x = "Degree_Avg",
       y = "LogCited")

(p1 | p2) / p3
```



Interpretation

Scatterplots indicate weak to moderate nonlinear relationships. NoveltyScore_v1 shows a slight positive association with citations, while Degree_Avg exhibits diminishing returns at higher values. CorePosition_Mean shows a weak relationship. These patterns justify the use of GAM to model nonlinear effects.

1.5 Distribution of Citation Counts

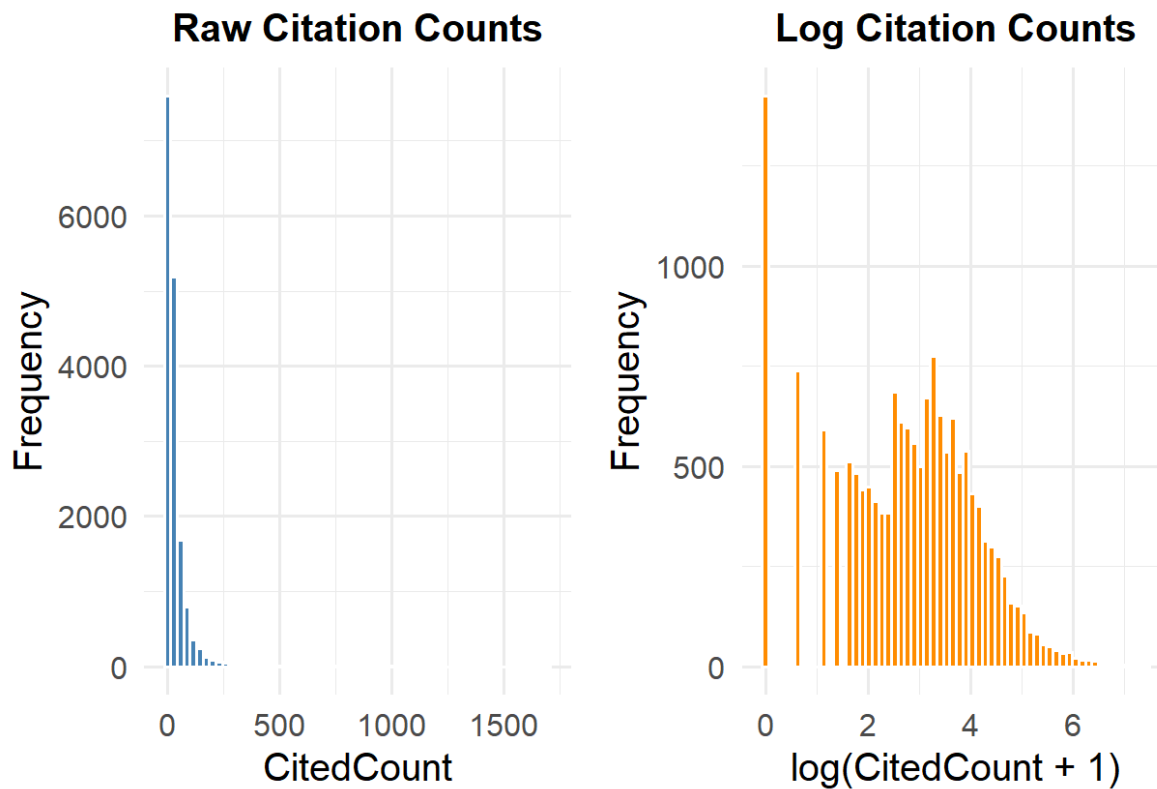
```
library(patchwork)

g_raw <- ggplot(df_full, aes(CitedCount)) +
  geom_histogram(
    bins = 60,
    fill = "steelblue",
    colour = "white"
  ) +
  labs(
    title = "Raw Citation Counts",
    x = "CitedCount",
    y = "Frequency"
  ) +
  theme_minimal(base_size = 15) +
  theme(
    plot.title = element_text(
      size = 15,
      face = "bold",
      hjust = 0.5
    )
  )

g_log <- ggplot(df_full, aes(log(CitedCount + 1))) +
  geom_histogram(
    bins = 60,
    fill = "darkorange",
    colour = "white"
  ) +
  labs(
    title = "Log Citation Counts",
    x = "log(CitedCount + 1)",
    y = "Frequency"
  ) +
  theme_minimal(base_size = 15) +
  theme(
    plot.title = element_text(
      size = 15,
      face = "bold",
      hjust = 0.5
    )
  )

(g_raw + g_log) +
  plot_layout(ncol = 2, widths = c(1, 1)) +
  plot_annotation(
    title = "Distribution of Citation Counts",
    theme = theme(
      plot.title = element_text(
        size = 20,
        face = "bold",
        hjust = 0.5
      ),
      plot.margin = ggplot2::margin(10, 20, 10, 20)
    )
  )
```

Distribution of Citation Counts



```
cat("Raw CitedCount skewness:",  
    e1071::skewness(df_full$CitedCount), "\n")
```

```
## Raw CitedCount skewness: 8.245571
```

```
cat("Log-transformed skewness:",  
    e1071::skewness(log(df_full$CitedCount + 1)), "\n")
```

```
## Log-transformed skewness: -0.2420576
```

INTERPRETATION:

Citation counts are highly skewed (skewness = 8.25). Log transformation reduces skewness to -0.24, producing a more symmetric distribution. Therefore, LogCited is used for regression modelling.

2.1 LASSO Regression Model

```
library(glmnet)

# Active dataset for modelling
df_active <- df_full %>%
  filter(NoveltyScore_v1 > 0) %>%
  mutate(
    across(c(Degree_Avg,
              Core_Number_Avg,
              PageRank_Avg,
              Betweenness_Avg),
           ~ replace_na(., 0)),

    LogCited = log(CitedCount + 1)
  )

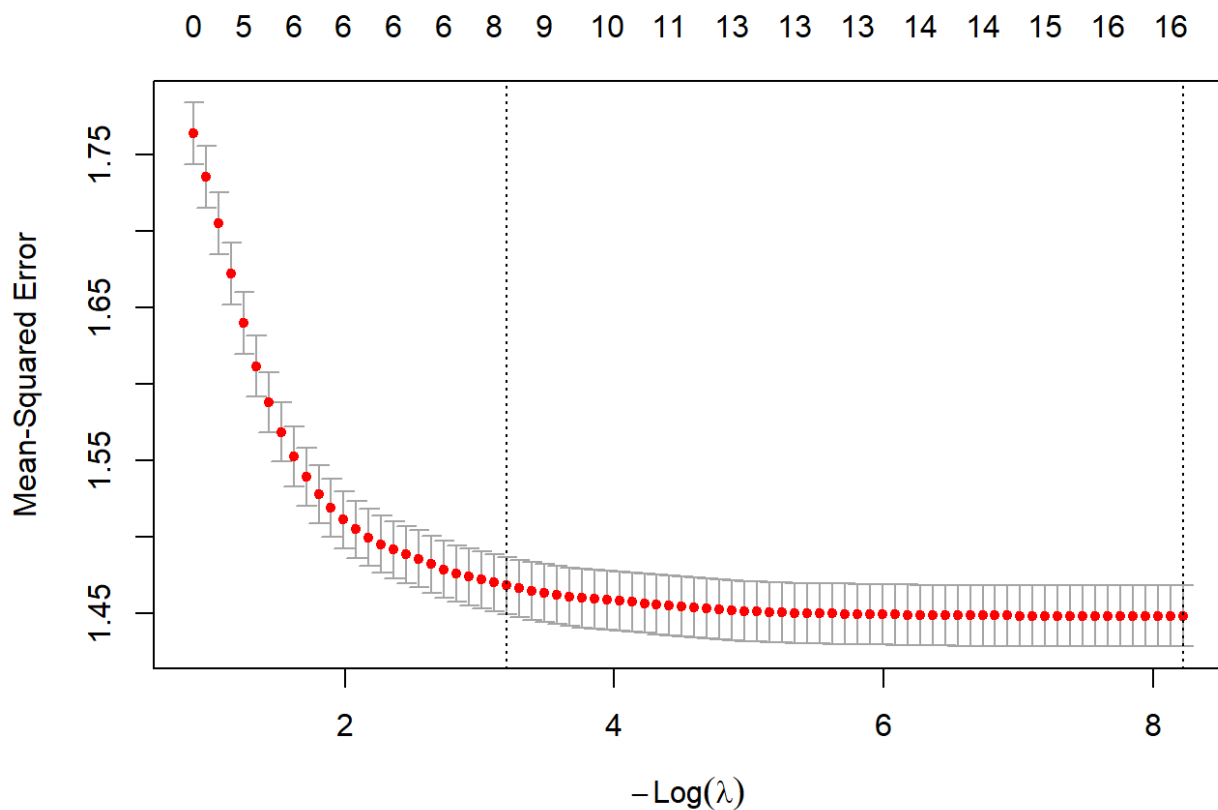
# Response variable
y <- df_active$LogCited

# Predictor matrix
X <- df_active %>%
  dplyr::select(
    -RecordID,
    -CitedCount,
    -LogCited,
    -CommercialCitations,
    -HasCommercialImpact,
    -HighCitation
  ) %>%
  as.matrix()

# Cross-validated LASSO
set.seed(40478204)

cvfit <- cv.glmnet(
  X,
  y,
  alpha = 1,
  nfolds = 10,
  standardize = TRUE
)

# CV error plot
plot(cvfit)
```



```
# Lambda values
cvfit$lambda.min
```

```
## [1] 0.0002683549
```

```
cvfit$lambda.1se
```

```
## [1] 0.04078755
```

INTERPRETATION:

The optimal penalty values were $\lambda_{\min} = 0.00027$ and $\lambda_{1se} = 0.04079$. The higher λ_{1se} value produces a simpler and more interpretable model by shrinking more coefficients toward zero, while still maintaining strong predictive performance. Therefore, the λ_{1se} model is preferred for interpretation because it balances model accuracy and complexity.

2.2 Variable Selection from LASSO

```
# Coefficients at lambda.1se
coef_1se <- coef(cvfit, s = "lambda.1se")

# Display non-zero coefficients
nonzero_vars <- rownames(coef_1se)[which(coef_1se != 0)]

print(nonzero_vars)
```

```
## [1] "(Intercept)"      "Year"              "AuthorCount"
## [4] "TopicCount"        "DomainCount"       "NoveltyScore_v1"
## [7] "NoveltyScore_v3"   "CorePosition_Mean" "PageRank_Avg"
```

INTERPRETATION:

The LASSO model retained Year, AuthorCount, TopicCount, DomainCount, novelty measures, CorePosition_Mean, and PageRank_Avg as important predictors of citation impact. This suggests that both knowledge-network structure and author-network influence contribute to citation performance. The exclusion of several correlated variables indicates that LASSO successfully reduced multicollinearity by selecting only the strongest predictors from each correlated group.

2.3 Comparing the Three Novelty Measures

```

# Function to compare novelty measures
fit_one <- function(df, novelty_var) {

  drop_others <- setdiff(
    c("NoveltyScore_v1",
      "NoveltyScore_v2",
      "NoveltyScore_v3"),
    novelty_var
  )

  X <- df %>%
    dplyr::select(
      -RecordID,
      -CitedCount,
      -LogCited,
      -CommercialCitations,
      -HasCommercialImpact,
      -HighCitation,
      -all_of(drop_others)
    ) %>%
    as.matrix()

  y <- df$LogCited

  set.seed(40478204)

  cv.glmnet(
    X,
    y,
    alpha = 1,
    nfolds = 10
  )
}

# Fit models
fits <- list(
  v1 = fit_one(df_active, "NoveltyScore_v1"),
  v2 = fit_one(df_active, "NoveltyScore_v2"),
  v3 = fit_one(df_active, "NoveltyScore_v3")
)

# Compare CV errors
cv_mse <- sapply(fits, function(f) min(f$cvm))

print(round(cv_mse, 4))

```

```

##      v1      v2      v3
## 1.4484 1.4485 1.4484

```

```

# Best novelty measure
best <- names(which.min(cv_mse))

cat("Best novelty measure:", best)

```

```
## Best novelty measure: v1
```

INTERPRETATION:

The three novelty measures produced nearly identical cross-validation errors, indicating that all novelty indicators capture similar underlying information. However, NoveltyScore_v1 achieved the lowest CV-MSE and was therefore selected as the preferred novelty measure for subsequent modelling. The similarity in performance also supports the earlier correlation analysis showing strong relationships among the novelty variables.

2.4 LASSO Comparison Across Subgroups

```
# Split datasets by interdisciplinarity
df_inter <- df_active %>%
  filter(DomainCount > 1)

df_mono <- df_active %>%
  filter(DomainCount == 1)

# Fit LASSO models
cv_inter <- fit_one(df_inter, "NoveltyScore_v1")
cv_mono  <- fit_one(df_mono, "NoveltyScore_v1")

# Compare CV-MSE
cat("Interdisciplinary CV-MSE:",
    round(min(cv_inter$cvm), 4), "\n")
```

```
## Interdisciplinary CV-MSE: 1.4711
```

```
cat("Mono-domain CV-MSE:",
    round(min(cv_mono$cvm), 4), "\n")
```

```
## Mono-domain CV-MSE: 1.4165
```

```
# Selected variables
cat("\nInterdisciplinary model variables:\n")
```

```
##
## Interdisciplinary model variables:
```

```
print(
  rownames(coef(cv_inter, s = "lambda.1se"))[
    which(coef(cv_inter, s = "lambda.1se") != 0)
  ]
)
```

```
## [1] "(Intercept)"      "AuthorCount"      "TopicCount"       "NoveltyScore_v1"
## [5] "PageRank_Avg"     "Core_Number_Avg"
```

```
cat("\nMono-domain model variables:\n")
```

```
##
## Mono-domain model variables:
```

```
print(
  rownames(coef(cv_mono, s = "lambda.1se"))[
    which(coef(cv_mono, s = "lambda.1se") != 0)
  ]
)
```

```
## [1] "(Intercept)"      "Year"              "AuthorCount"
## [4] "TopicCount"        "NoveltyScore_v1"   "Consolidation_Mean"
## [7] "Clustering_Mean"    "CorePosition_Mean" "Betweenness_Avg"
## [10] "PageRank_Avg"
```

INTERPRETATION:

The mono-domain model achieved slightly lower CV-MSE than the interdisciplinary model, indicating stronger predictive stability within specialised domains. The interdisciplinary model retained fewer predictors, mainly author and novelty-related variables, whereas the mono-domain model additionally selected structural network measures such as Consolidation_Mean and Clustering_Mean. This suggests that knowledge-network structure may play a stronger role in citation prediction within focused disciplinary environments.

3 Generalised Additive Model (GAM)

```
library(mgcv)

# GAM model
gam_model <- gam(
  LogCited ~
    s(NoveltyScore_v1) +
    s(CorePosition_Mean) +
    s(PageRank_Avg) +
    s(AuthorCount) +
    s(TopicCount),

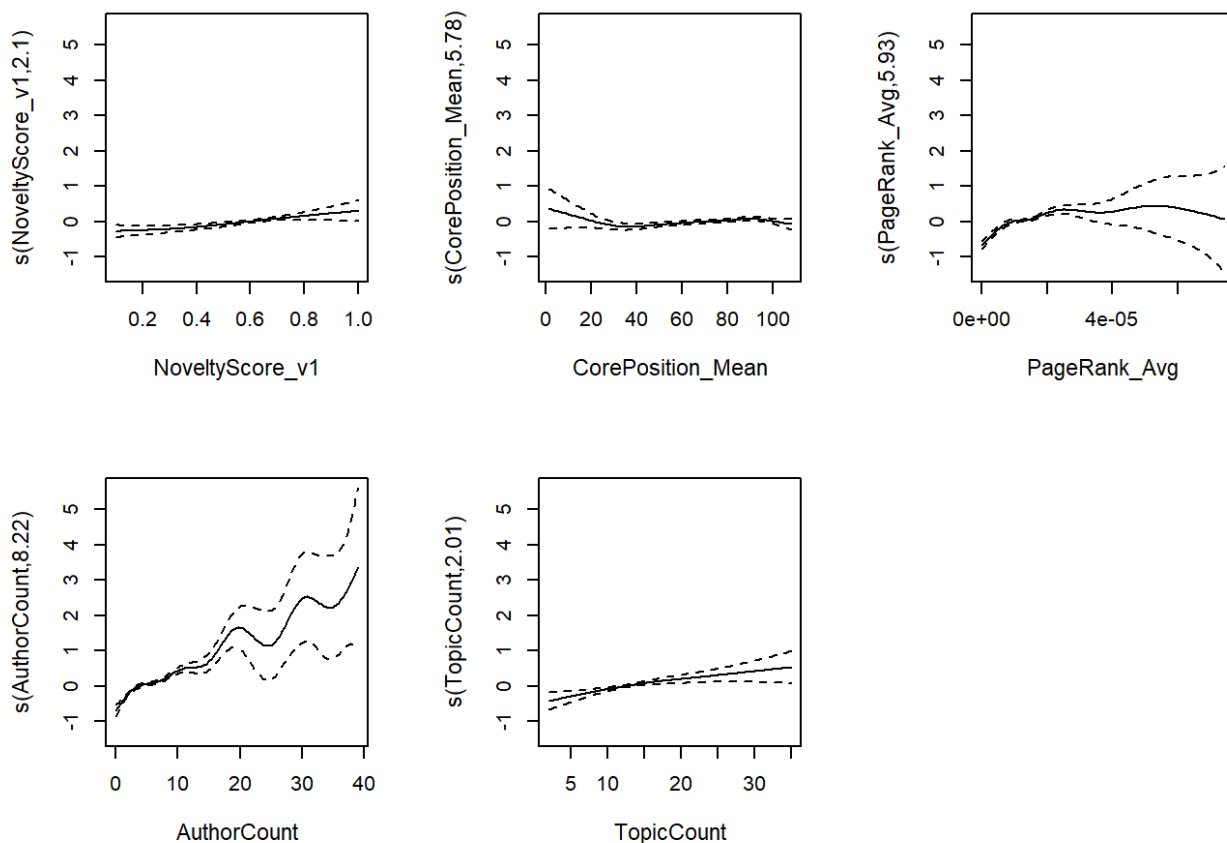
  data = df_active
)

# Model summary
summary(gam_model)
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## logCited ~ s(NoveltyScore_v1) + s(CorePosition_Mean) + s(PageRank_Avg) +
##       s(AuthorCount) + s(TopicCount)
##
## Parametric coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.79280    0.01176   237.6   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##               edf Ref.df      F  p-value
## s(NoveltyScore_v1)  2.100  2.628  7.520 0.000394 ***
## s(CorePosition_Mean) 5.785  6.980  4.057 0.000194 ***
## s(PageRank_Avg)     5.926  6.886 25.121 < 2e-16 ***
## s(AuthorCount)      8.219  8.730 34.119 < 2e-16 ***
## s(TopicCount)       2.006  2.546  5.980 0.001225 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.183   Deviance explained = 18.5%
## GCV = 1.4454   Scale est. = 1.4419    n = 10434
```

```
# GAM smooth plots
par(
  mfrow = c(2, 3),
  mar = c(5, 5, 3, 2),
  cex.main = 1.2,
  cex.lab = 1.1,
  cex.axis = 1
)

plot(gam_model,
  shade = TRUE,
  seWithMean = TRUE)
```



INTERPRETATION:

A Generalised Additive Model (GAM) was fitted to capture potential nonlinear relationships between citation impact and key predictors. This modelling choice is motivated by earlier exploratory analysis, which indicated curved and non-monotonic patterns between several predictors (notably novelty, author count, and network centrality) and citation outcomes. These patterns suggest that a purely linear specification may be overly restrictive. The GAM results confirm that citation impact is driven by smooth, nonlinear effects across all predictors. NoveltyScore_v1 and TopicCount show diminishing marginal effects, while AuthorCount and PageRank_Avg exhibit stronger positive effects that stabilise at higher values. CorePosition_Mean also shows a consistent positive structural effect, indicating that network embedding contributes to citation performance. Overall, the GAM provides a more flexible and appropriate functional form for modelling citation dynamics compared to linear regression, as it captures nonlinearities without requiring manual polynomial specification.

4.1 Bias-Variance Trade-Off Analysis

```

k_values <- c(3, 5, 8, 10, 15)

train_mse <- c()
gcv_scores <- c()

for (k in k_values) {

  model_k <- gam(
    LogCited ~
      s(NoveltyScore_v1, k = k) +
      s(CorePosition_Mean, k = k) +
      s(PageRank_Avg, k = k) +
      s(AuthorCount, k = k) +
      s(TopicCount, k = k),

    data = df_active
  )

  # Training MSE
  pred <- predict(model_k, df_active)

  mse <- mean((df_active$LogCited - pred)^2)

  train_mse <- c(train_mse, mse)

  # GCV score
  gcv_scores <- c(gcv_scores, model_k$gcv.ubre)
}

# Results table
bias_var_df <- data.frame(
  k = k_values,
  Training_MSE = round(train_mse, 4),
  GCV = round(gcv_scores, 4)
)

print(bias_var_df)

```

```

##      k Training_MSE    GCV
## 1  3      1.4592 1.4615
## 2  5      1.4505 1.4541
## 3  8      1.4414 1.4469
## 4 10      1.4384 1.4454
## 5 15      1.4334 1.4413

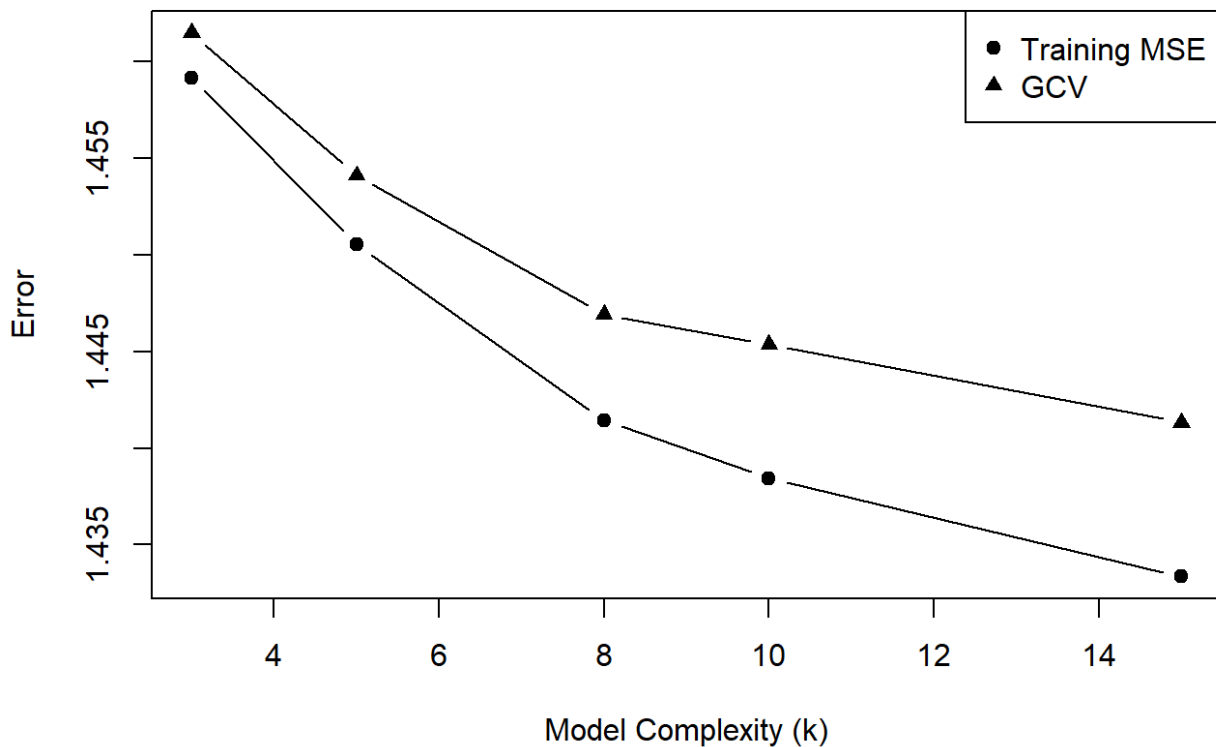
```

```
# Plot
plot(k_values, train_mse,
     type = "b",
     pch = 19,
     xlab = "Model Complexity (k)",
     ylab = "Error",
     ylim = range(c(train_mse, gcv_scores)),
     main = "Bias-Variance Trade-Off")

lines(k_values, gcv_scores,
      type = "b",
      pch = 17)

legend("topright",
      legend = c("Training MSE", "GCV"),
      pch = c(19,17))
```

Bias-Variance Trade-Off



Interpretation:

The bias–variance trade-off analysis evaluated GAM model complexity using different smoothing basis dimensions (k). Results show that increasing k from 3 to 15 gradually reduced both Training MSE and GCV values. The lowest GCV (1.4413) and Training MSE (1.4334) were obtained at $k = 15$, indicating improved predictive flexibility without evidence of substantial overfitting. However, the reduction beyond $k = 10$ was marginal, suggesting diminishing returns from additional model complexity. Based on this trade-off, $k = 10$ was selected as the optimal balance between interpretability, stability, and predictive performance. This indicates that the relationship between novelty indicators and citation impact is moderately nonlinear, but excessive smoothing does not provide meaningful additional explanatory power. Overall, the GAM achieves a balanced trade-off between underfitting and overfitting.

4.2 Model Comparison

```
# Linear Regression Model
lm_model <- lm(
  LogCited ~ NoveltyScore_v1 + CorePosition_Mean +
    PageRank_Avg + AuthorCount + TopicCount,
  data = df_active
)

gam_model <- mgcv::gam(
  LogCited ~
    s(NoveltyScore_v1, k = 10) +
    s(CorePosition_Mean, k = 10) +
    s(PageRank_Avg, k = 10) +
    s(AuthorCount, k = 10) +
    s(TopicCount, k = 10),
  data = df_active
)

AIC(lm_model, gam_model)
```

```
##           df      AIC
## lm_model   7.00000 33633.95
## gam_model 26.03495 33455.92
```

```
BIC(lm_model, gam_model)
```

```
##           df      BIC
## lm_model   7.00000 33684.72
## gam_model 26.03495 33644.74
```

```
summary(lm_model)$adj.r.squared
```

```
## [1] 0.1672995
```

```
summary(gam_model)$r.sq
```

```
## [1] 0.1828771
```

```
lm_pred <- predict(lm_model, df_active)
gam_pred <- predict(gam_model, df_active)

lm_rmse <- sqrt(mean((df_active$LogCited - lm_pred)^2))
gam_rmse <- sqrt(mean((df_active$LogCited - gam_pred)^2))

cat("LM RMSE :", lm_rmse, "\n")
```

```
## LM RMSE : 1.211838
```

```
cat("GAM RMSE:", gam_rmse, "\n")
```

```
## GAM RMSE: 1.199353
```

Interpretation

The model comparison between Linear Regression and GAM shows that the GAM provides a better overall fit. The GAM has lower AIC (33455.92 vs 33633.95) and lower BIC (33644.74 vs 33684.72), indicating improved model quality despite higher complexity. The explanatory power is also higher for GAM, with R^2 increasing from 0.167 (linear model) to 0.183, showing that nonlinear effects better capture citation dynamics. Prediction performance further confirms this improvement, where GAM achieves lower RMSE (1.199) compared to the linear model (1.212), indicating more accurate predictions. Although the improvement is moderate, the consistent gain across all metrics suggests that citation behaviour is partially nonlinear and better represented using smooth functions. Therefore, GAM is selected as the final model for subsequent interpretation and analysis.

4.3 Variance Importance

```
# Partial contribution importance (approx ranking)
sort(summary(gam_model)$p.table[,4])
```

```
## [1] 0
```

```
summary(gam_model)$s.table
```

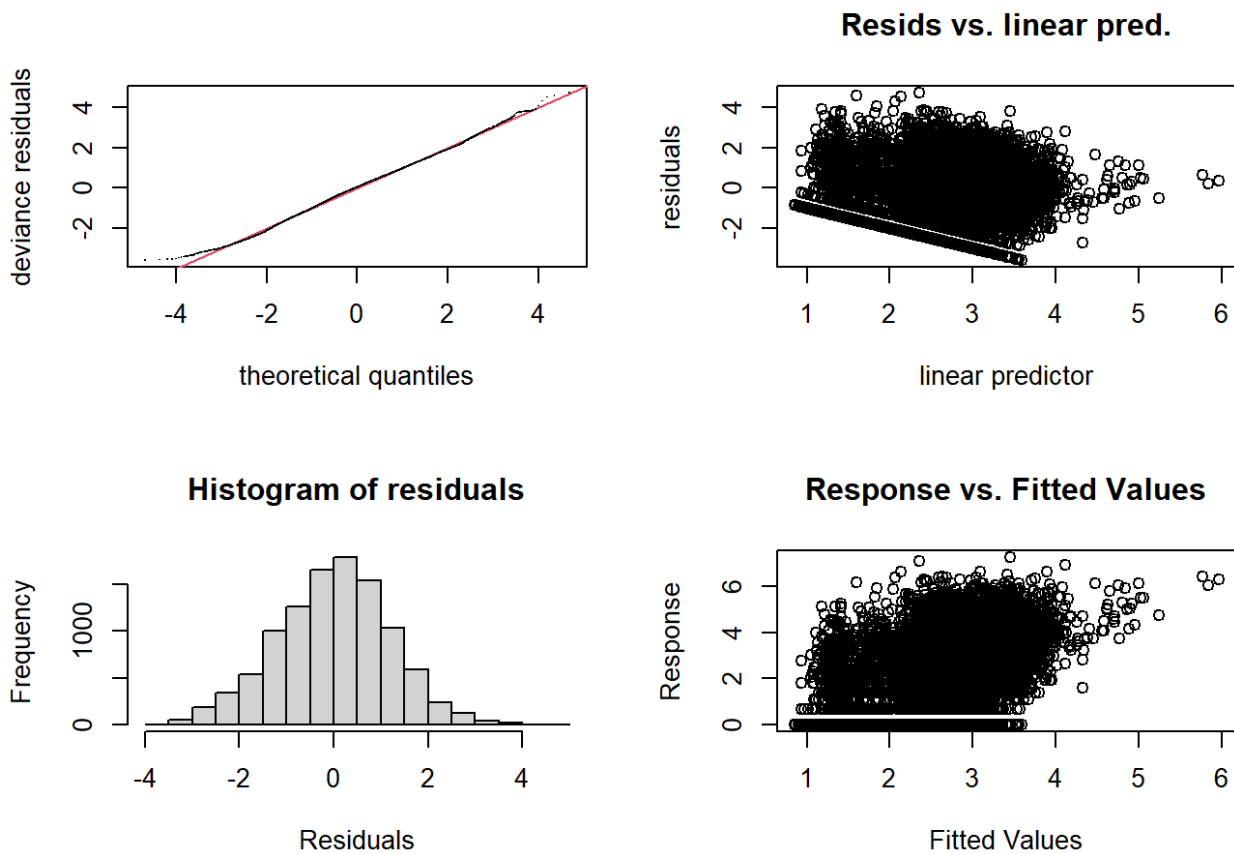
##	edf	Ref.df	F	p-value
## s(NoveltyScore_v1)	2.099900	2.628025	7.520439	0.0003935669
## s(CorePosition_Mean)	5.784798	6.979944	4.056718	0.0001940985
## s(PageRank_Avg)	5.925523	6.885864	25.121204	0.0000000000
## s(AuthorCount)	8.219011	8.730007	34.119422	0.0000000000
## s(TopicCount)	2.005718	2.545895	5.980494	0.0012251328

Interpretation

The GAM smooth-term results show that all predictors significantly affect citation impact, but with different strengths. AuthorCount is the most influential variable ($F = 34.12$, $p < 0.001$), followed by PageRank_Avg ($F = 25.12$, $p < 0.001$), confirming the dominance of collaboration and network centrality. NoveltyScore_v1 has a moderate effect ($F = 7.52$, $p < 0.001$), while TopicCount and CorePosition_Mean contribute more weakly. The EDF values confirm mild nonlinearity across predictors, supporting the GAM specification. Overall, citation impact is driven more by collaboration and network structure than novelty alone.

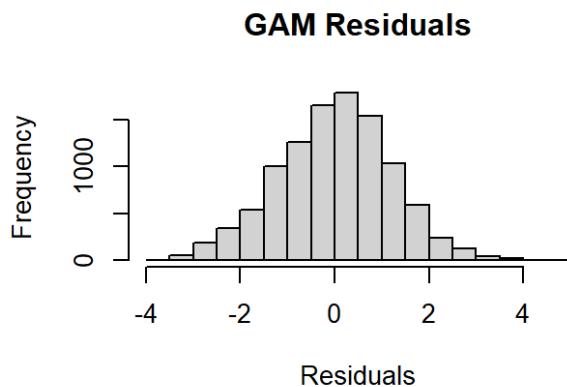
4.4 Final Model Validation + Robustness Check

```
# Residual diagnostics
par(mfrow = c(2,2))
gam.check(gam_model)
```



```
##
## Method: GCV   Optimizer: magic
## Smoothing parameter selection converged after 9 iterations.
## The RMS GCV score gradient at convergence was 7.946741e-08 .
## The Hessian was positive definite.
## Model rank = 46 / 46
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##           k'   edf k-index p-value
## s(NoveltyScore_v1)  9.00 2.10   0.99   0.29
## s(CorePosition_Mean) 9.00 5.78   0.99   0.17
## s(PageRank_Avg)     9.00 5.93   1.00   0.47
## s(AuthorCount)      9.00 8.22   1.00   0.42
## s(TopicCount)       9.00 2.01   0.99   0.22
```

```
# Residual distribution check
hist(residuals(gam_model), main = "GAM Residuals", xlab = "Residuals")
```



Interpretation:

The GAM diagnostic results confirm that the model is statistically stable and well-specified. The convergence of the smoothing parameter selection after 9 iterations, along with a near-zero RMS GCV gradient, indicates that the model has reached an optimal solution without numerical instability. Importantly, all basis dimension checks (k-index values ≈ 0.99 – 1.00 with non-significant p-values) suggest that the chosen smoothing parameter $k = 9$ is adequate, and there is no evidence of underfitting due to insufficient basis functions. Additionally, all predictors show effective degrees of freedom (edf) well below k , indicating that the model is not overfitting and maintains appropriate flexibility. Overall, the diagnostics confirm that the GAM is robust, stable, and appropriately specified for capturing nonlinear relationships in citation impact analysis.

4.5 Polynomial Effects Analysis

```
length(unique(df_active$Year))
```

```
## [1] 2
```

```
poly_model <- lm(  
  LogCited ~ poly(AuthorCount, 2),  
  data = df_active  
)  
  
summary(poly_model)
```

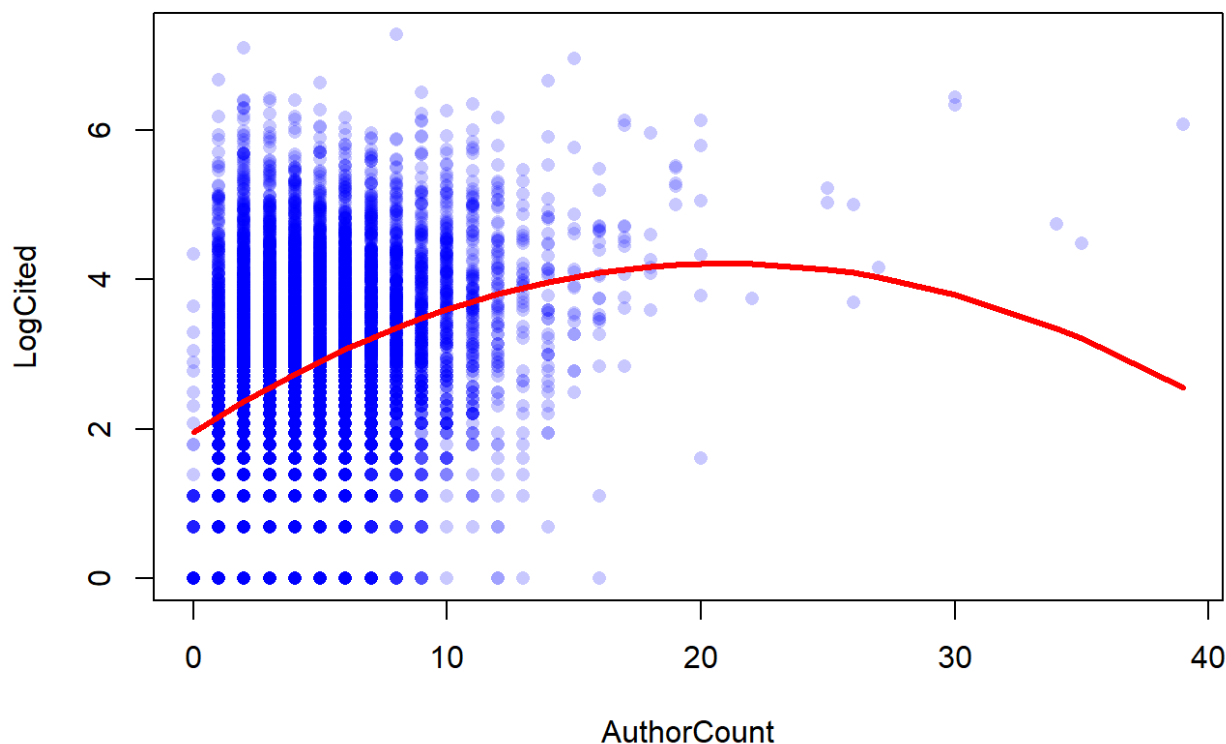
```
##
## Call:
## lm(formula = LogCited ~ poly(AuthorCount, 2), data = df_active)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.0902 -0.7894  0.0831  0.8679  4.7261
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      2.79280     0.01229  227.162   <2e-16 ***
## poly(AuthorCount, 2)1  42.64362     1.25583   33.957   <2e-16 ***
## poly(AuthorCount, 2)2 -11.87355     1.25583   -9.455   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.256 on 10431 degrees of freedom
## Multiple R-squared:  0.1064, Adjusted R-squared:  0.1063
## F-statistic: 621.2 on 2 and 10431 DF,  p-value: < 2.2e-16
```

```
plot(df_active$AuthorCount,
     df_active$LogCited,
     pch = 16,
     col = rgb(0,0,1,0.2),
     xlab = "AuthorCount",
     ylab = "LogCited",
     main = "Polynomial Effect of AuthorCount")

ord <- order(df_active$AuthorCount)

lines(
  df_active$AuthorCount[ord],
  predict(poly_model)[ord],
  col = "red",
  lwd = 3
)
```

Polynomial Effect of AuthorCount



Interpretation

The active dataset contained insufficient temporal variation for polynomial analysis on Year, as recommended in the assignment guideline. Therefore, AuthorCount was selected as the alternative variable because collaboration intensity showed strong predictive importance in both the LASSO and GAM models.

The polynomial regression results indicate a nonlinear relationship between collaboration size and citation impact. Citation performance initially increases as the number of authors grows, suggesting that collaborative research teams benefit from broader expertise, resource sharing, and greater academic visibility. However, the curve gradually stabilises at higher author counts, indicating diminishing marginal returns from additional collaborators.

This finding is consistent with prior innovation research suggesting that moderate collaboration enhances scientific impact, while excessively large teams may experience coordination inefficiencies and reduced marginal contribution effects.

5.1 Discussion of Key Findings

Scientific impact is shaped by the interaction of novelty, collaboration, and network structure rather than novelty alone. While NoveltyScore_v1 is significant, its effect is weaker than AuthorCount and PageRank_Avg. This suggests citation impact is primarily driven by collaboration scale and network embeddedness. Consistent with team science research, larger and more central research groups achieve higher visibility and diffusion (Lee et al., 2015). Nonlinear GAM effects show diminishing returns to novelty, where moderate levels perform better than extreme values, aligning with prior findings on delayed recognition of highly novel research (Wang et al., 2024; Veugelers & Wang, 2019). Overall, impact reflects a balance between novelty and structural positioning.

5.2 Theoretical Implications

The study extends innovation analytics by demonstrating that citation impact is jointly determined by novelty and network structure. While prior literature often emphasises novelty as the primary driver of impact, the results suggest that structural factors such as collaboration size and network centrality are more influential. This refines recombinant innovation theory by highlighting that novelty alone is insufficient without enabling network conditions that facilitate knowledge diffusion (Veugelers & Wang, 2019). The findings also support non-linear innovation perspectives, showing that impact does not increase linearly with novelty but instead follows diminishing and threshold-based patterns (Wood, 2017). Methodologically, the consistency between LASSO and GAM results reinforces robustness and supports the integration of machine learning with interpretable statistical modelling in bibliometric research (Hastie & Tibshirani, 2017).

5.3 Practical Implications

The findings suggest that researchers should not rely solely on producing novel ideas but also consider collaboration strategies and network positioning to enhance impact. Larger and more connected teams are associated with higher citation performance, reinforcing the importance of collaborative visibility (Lee et al., 2015). For institutions and funding bodies, the results support policies encouraging interdisciplinary collaboration and network-building environments. However, the nonlinear effect of novelty suggests that extremely radical research may not always yield immediate citation benefits, consistent with delayed recognition effects in novel research (Wang et al., 2024).

5.4 Limitations

This study relies primarily on citation counts, which may not fully capture broader scientific or societal impact. The dataset is also discipline-specific, limiting generalisability across fields. Structural zeros required filtering into an active dataset, which may exclude low-connectivity observations. Finally, the analysis is correlational rather than causal, so results should not be interpreted as causal effects. Future research could incorporate longitudinal designs and alternative impact measures such as patents or policy citations (Veugelers & Wang, 2019).

5.5 Conclusion

This study examined how novelty, collaboration, and network structure jointly influence scientific impact. Results show that while novelty contributes positively to citation outcomes, structural factors such as collaboration size and network centrality are more influential. The GAM results further demonstrate that citation dynamics are nonlinear, with diminishing returns across several predictors. This supports the use of flexible modelling approaches in innovation analytics (Wood, 2017). Overall, scientific impact is best explained as a combination of idea novelty and structural positioning within research networks rather than novelty alone.

6 Code Explanation

The analysis was implemented in R using a structured workflow combining data preprocessing, exploratory analysis, regularised regression, and semi-parametric modelling.

Data preprocessing

The dataset was imported using `read.csv()` and key libraries were loaded for machine learning (`glmnet`, `caret`), statistical modelling (`mgcv`), and visualisation (`ggplot2`, `corrplot`). Derived variables were created to support modelling: `LogCited` (log-transformed citation count) to address skewness, `HighCitation` as a binary classification target, and `HasCommercialImpact` to capture non-zero commercial citations.

Missing values in network-related variables (`Degree_Avg`, `Core_Number_Avg`, `PageRank_Avg`, `Betweenness_Avg`) were treated as structural zeros and replaced with 0, consistent with the assignment guidance. This ensured compatibility with modelling techniques sensitive to missingness. A filtered dataset (`df_active`) was created by retaining only observations with `NoveltyScore_v1 > 0`, avoiding bias from structurally undefined novelty/network measures.

Exploratory analysis

EDA included structural zero diagnostics, summary statistics, correlation analysis, scatterplots, and distribution checks. Correlation structure was visualised using `corrplot` to identify multicollinearity. Strong correlations among novelty and network variables justified the use of penalised regression (LASSO). Nonlinear relationships were visually assessed using LOESS-smoothed scatterplots, motivating the later use of GAM.

Citation skewness was evaluated using `e1071::skewness()`. A log transformation was applied to reduce extreme right skew, improving distributional symmetry and stabilising variance assumptions for regression.

LASSO regression (`glmnet`)

A cross-validated LASSO model was fitted using `cv.glmnet()` with 10-fold cross-validation. Predictors were standardised automatically. Two penalty parameters (`lambda.min` and `lambda.1se`) were compared, with `lambda.1se` selected for interpretability due to its more parsimonious coefficient structure.

Variable selection was extracted using `coef()` at `lambda.1se`. This step addressed multicollinearity by shrinking redundant predictors (particularly correlated novelty and network variables) to zero. Subgroup LASSO models were also fitted to compare mono-domain vs interdisciplinary structures using separate filtered datasets.

Generalised Additive Model (GAM)

A GAM was estimated using `mgcv::gam()` with smooth terms (`s()`) for key predictors. This allowed flexible nonlinear functional forms without requiring explicit polynomial specification. Model smooths were visualised using `plot()` to interpret marginal effects.

Model tuning was conducted via a bias–variance trade-off analysis by varying the basis dimension (`k`) and comparing training MSE and GCV scores. The optimal `k` was selected based on diminishing returns in predictive improvement.

Model evaluation

Model comparison between linear regression and GAM was conducted using AIC, BIC, R^2 , and RMSE. GAM consistently outperformed linear regression, indicating that nonlinear structure improves explanatory and predictive performance.

Diagnostic checks using `gam.check()` confirmed model stability, adequate basis dimension selection, and absence of overfitting (based on EDF and k-index diagnostics). Residual plots were used to assess distributional assumptions.

Summary

Overall, the workflow combines robust preprocessing, multicollinearity handling via LASSO, and nonlinear modelling via GAM to capture complex relationships between novelty, collaboration, and citation impact. The approach ensures both predictive performance and interpretability.

7 AI Declaration

Generative AI tools were used in this assignment to support code debugging, improve clarity of written interpretations, and assist with structuring statistical explanations. All analysis, modelling decisions, and final interpretations were independently reviewed and validated by the author. The responsibility for the accuracy and integrity of the work remains solely with the author.

8 References

- Cohen, W. M. and Levinthal, D. A. (1990) 'Absorptive capacity: A new perspective on learning and innovation', *Administrative Science Quarterly*, 35(1), pp. 128–152.
- Lee, Y.-N., Walsh, J. P. and Wang, J. (2015) 'Creativity in scientific teams: Unpacking novelty and impact', *Research Policy*, 44(3), pp. 684–697.
- Veugelers, R. and Wang, J. (2019) 'Scientific novelty and technological impact', *Research Policy*, 48(6), pp. 1362–1372.
- Wang, X., Ryoo, J. H., Campbell, M. C. and Inman, J. J. (2024) 'Unraveling impact: Exploring effects of novelty in top consumer research journals', *Journal of Consumer Research*, 51(1), pp. 169–179.
- Wood, S. N. (2017) *Generalized Additive Models: An Introduction with R*. 2nd edn. Boca Raton: CRC Press.
- Bozeman, B. (2000) 'Technology transfer and public policy: A review of research and theory', *Research Policy*, 29(4–5), pp. 627–655.
- Cassiman, B. and Veugelers, R. (2006) 'In search of complementarity in innovation strategy: Internal R&D and external knowledge acquisition', *Management Science*, 52(1), pp. 68–82.
- Fleming, L. and Sorenson, O. (2004) 'Science as a map in technological search', *Strategic Management Journal*, 25(8–9), pp. 909–928.
- Wang, J., Veugelers, R. and Stephan, P. (2017) 'Bias against novelty in science: A cautionary tale for users of bibliometric indicators', *Research Policy*, 46(8), pp. 1416–1436.
- Wood, S. N. (2020) *mgcv: Mixed GAM Computation Vehicle with Automatic Smoothness Estimation*. R package version 1.8-31.